

SERGEY CHERNYAKOV



Senior Data Engineer · Python Data Pipelines · ML / AI Data Platform

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Age: 47 · Male

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Location: Wrocław, Poland · Remote / EU · B2B available immediately

PROFESSIONAL SUMMARY

Senior engineer with **20+ years** in commercial software and a current focus on **data engineering, Python data pipelines, ML / AI data platforms and production observability**. Designs and operates **ETL / streaming pipelines** that move large, heterogeneous data into analytics, ML and LLM workloads — with strict attention to **schema versioning, data quality, governance and reproducibility**.

Hands-on with **Python (FastAPI, async I/O), PostgreSQL, Kafka, MongoDB, Redis, Docker, GitHub Actions CI/CD**, AI-assisted SDLC and ML stacks (**XGBoost, scikit-learn**). Familiar with **Databricks-adjacent patterns** — PySpark, Delta-style schema-versioned data, MLflow concepts, parquet, columnar storage, MEV-aware deterministic pipelines on financial data. Built **drift detection, feature-stability protocols and backtest-live parity** mechanisms used in production.

Available as **B2B contractor** (Wrocław-based, fully remote OK), starting ASAP.

KEY SKILLS

- **Data engineering:** ETL / ELT pipelines · streaming (Kafka) · batch & micro-batch · schema-versioned datasets · idempotent jobs · backfill strategies · data-quality checks · observability and lineage
- **Languages / runtime:** **Python** (primary — FastAPI / Django / Flask, asyncio, multiprocessing) · SQL · TypeScript / Node.js · Ruby
- **Databricks / big-data adjacent:** PySpark concepts · Delta-style schema versioning · parquet / columnar formats · MLflow patterns · feature stores · dataset reproducibility
- **Storage & messaging:** **PostgreSQL** (async drivers: asyncpg / psycopg2) · MySQL · MongoDB · Redis · Apache Kafka · WebSockets data feeds
- **Cloud / DevOps:** Docker · GitHub Actions CI/CD · GCP basics · bare-metal Nginx / PHP-FPM / systemd · environment-aware deployments
- **Data governance & security:** schema versioning · feature-stability protocols (5-seed) · config-hash drift detection (runtime WARN) · daily honesty checks (shuffled-label, prior-drift, economic backtest) · secrets management · access boundaries
- **ML / analytics:** **XGBoost** · scikit-learn · feature engineering · quantile regression · survival / hazard models · backtest-live parity protocols
- **AI / LLM data workflows:** OpenAI API · Anthropic Claude · LangChain / LangGraph · MCP servers + SSE · vector DBs / RAG · document AI (OCR, layout-preserving translation, classification)

PROFESSIONAL EXPERIENCE

AI Engineering Lead / Data Pipelines Architect — Sabermine (United Kingdom, Remote)

November 2024 – November 2025

Led data and AI engineering for internal and client systems.

- Designed and operated Python-based **ETL and automation pipelines** for data ingestion, transformation and downstream analytics across heterogeneous sources
- Built **internal tooling for data quality, observability and reproducibility** of AI / ML pipelines — schema checks, lineage, drift detection
- Backend services on **Python, FastAPI, PostgreSQL**; Docker-based deployment, CI/CD via GitHub Actions
- Integrated **OpenAI APIs / LLM workflows** as data-enrichment steps in pipelines (OCR, document classification, structured-data extraction)
- Built **multi-agent AI systems** for automation, QA and data documentation
- Improved development speed and operational efficiency **5–10x** via AI-assisted SDLC

Senior Python / Data / Backend Engineer — Upwork (International Clients, Remote)

June 2012 – June 2025 (Top Rated)

18,000+ hours · 71+ contracts · 98% Job Success. Most data- and pipeline-relevant recent work:

  **Production ML pipelines with strict data quality**

- **Solana Pulse Bot** (`solana_pulse_bot`) — real-time observer for Solana memecoin launchpads. Two-phase scoring pipeline: fast filter at T+5s + full analysis at T+90s. **82 features per token (schema v21)**, **8 XGBoost ML heads** (entry main / entry @T+30 / entry regression / entry-timing 3-class / survival hazard / SL+TP+max_hold quantile heads). **100% backtest-live decision parity** verified on 300+ tokens. Daily 9 honesty checks (shuffled labels, prior drift, economic backtest), **5-seed feature-stability protocol**, **config-hash drift WARN at runtime**, schema versioning. Stack: Python, **asyncpg** + **psycopg2**, Streamlit dashboards, Helius API, WebSockets.

🔗 ETL & document data pipelines

- **Medical document translation pipelines** (`umetex`) — Django + ChatGPT translation of medical PDFs / Word / images, **layout-preserving** (tables, images), Cyrillic font handling, predefined-phrase consistency tables, IP-based history, async background processing.
- **Equipment recognition pipelines** (`tagman`) — GPT-Vision image classification + **structured metadata extraction** (serial numbers, equipment labels), Excel report export, parametric reproducibility.

🔗 LLM / agent data infrastructure

- **AQ AI Agent** (`aq_ai_agent`) — FastAPI + Model Context Protocol via Server-Sent Events; wraps legacy Aquarium SOAP CRM as agent-callable MCP tools. Pattern for exposing legacy enterprise data into modern AI / agent pipelines.
- **Weather MCP Agent** (`weather_mcp_agent`) — clean FastAPI + MCP SSE reference implementation.
- **AI-driven SDLC tooling** (`blank_python_project`) — Claude Code agents for automated planner / architect / coder / reviewer / QA workflows; pre-commit (isort / black / ruff / pylint), log rotation, type hints, reproducible Python project template.

🔗 Sentiment / analytics pipelines

- **Auto Sentiment** (`auto_sentiment`) — sentiment analysis on car-dealership customer comments; Excel-based business analytics ETL.

🔗 Content / streaming automation

- **Pauchok** (private) — automated content publishing pipeline to Yandex.Dzen with ChatGPT image generation, RSS-driven scheduling, Playwright browser automation.
- **TG Promoter** (`tgwriterbot`) — multi-account rotation, anti-ban logic (daily / hourly limits, cooldowns), Claude CLI integration for text generation.

Senior Fullstack Engineer — Fit2you (Italy, insurance)

April 2022 – January 2024

- Internal enterprise systems (dealer / vendor / client portals) backed by **PostgreSQL** data layer
- Migrated all company applications from **Rails 4.1** → **Rails 7.1** (5 legacy apps), SonarQube quality gates, 80% test coverage
- Supported and developed **all 8 of the company's applications** end-to-end — feature work, maintenance and incident fixes; one of them was a **Node.js + React** app, the rest on Ruby on Rails
- Stack: Ruby on Rails, Node.js / Express, TypeScript, PostgreSQL

Earlier roles — Clickfunnels (USA), Lawdingo (USA), Gleam.io (Australia), Skykillers Digital Agency, ITC Publishing, RBC-Ukraine

2004 – 2022

Senior backend engineering for high-traffic media, advertising platforms (Colgate, Danone, Nicorette, Visine, Davidoff, Johnson's Baby), payment platforms (Lawdingo: Stripe + Twilio + Sidekiq), high-volume marketing SaaS (Clickfunnels: Rails + MySQL + **Kafka** + Redis + 12+ integrations), SaaS competitions (Gleam.io: Rails + Angular + social APIs).

Reference from Ryan Montgomery (COO, ClickFunnels): *"Seriously one of the best hires we've ever made on Upwork. Sergey has continued to exceed our expectations."*

SELECTED OPEN-SOURCE / DATA PROJECTS

- **solana_pulse_bot** — Real-time ML pipeline with 8 XGBoost heads, 100% backtest-live parity, drift detection, schema versioning. github.com/sergeychernyakov/solana_pulse_bot
- **umetex** — Django + ChatGPT medical document translation pipeline with layout preservation. github.com/sergeychernyakov/umetex
- **aq_ai_agent** — FastAPI + MCP + SOAP-to-AI data bridge. github.com/sergeychernyakov/aq_ai_agent
- **weather_mcp_agent** — Clean MCP SSE reference for agent-data integration. github.com/sergeychernyakov/weather_mcp_agent
- **aihelper** — Multi-modal Telegram bots with OpenAI + payments + i18n. github.com/sergeychernyakov/aihelper
- **tgwriterbot** — TG automation with Claude CLI + anti-ban logic. github.com/sergeychernyakov/tgwriterbot
- **blank_python_project** — Python template with Claude Code SDLC agents. github.com/sergeychernyakov/blank_python_project

Private: *arbitraflashbot_v2* (DEX arbitrage with ML + flashloans), *pauchok* (Yandex.Dzen auto-publish pipeline), *tagman* (GPT-Vision OCR data extraction), *instreamatic* (WhatsApp + GPT bot pipeline), *frisson* (meter-reading neural net), *surveye* (video-analytics pipeline).

EDUCATION

Bachelor's Degree — Don State Technical University, 1999

LANGUAGES

Ukrainian — Native · English — C1 (Fluent professional) · Polish — Learning (A1)

ENGAGEMENT

- **B2B contractor** (own activity) — invoicing via Deel.com or direct
- **Format:** Fully remote (preferred) · Hybrid OK (1–2 days/week, Wrocław area)
- **Location:** Wrocław, Poland
- **Notice period:** Immediate / ~1 month
- Strong production / operational mindset — data quality, observability, governance and reproducibility as defaults